

The Bioprinter Project

A public bacterial-culture printing system — system overview, build documentation, and current status

1. Overview

This project turns bacteria into a public art medium: a modified 3D printer deposits a bacterial-alginate ink onto agar plates, following the outline of a user-submitted image, and the resulting culture grows into a visible, colored print over 12–24 hours. The system is being developed into a public, passcode-gated website (bioprinter.com) where anyone can upload a design, preview a traced version of it, and add it to a print queue; someone on-site then prepares the plate and runs the print.

This report documents how the system works end to end — the web app, the printer hardware, materials preparation, and the print procedure — along with what has been tested, what's working, and what's still in progress.

2. System Architecture

The pipeline has four stages:

- **Web app (Flask):** a local Flask application (app.py, line_trace_gcode.py, templates/index.html, templates/result.html) handles image upload, tracing, and gcode generation. It's being extended into the public passcode-gated site described in Section 3.
- **Image-to-gcode tracing:** uploaded images are converted to line paths using one of three modes — edge trace, skeletonize, or a hybrid of both (color fill is not yet built). Paths are scaled, offset, and hard-clipped to a 39.5 mm print-area radius so generated gcode never asks the printer to move outside the intended area.
- **Printer control (Klipper / Moonraker):** the printer runs Klipper firmware with a Moonraker API layer (Mainsail/Fluidd front end), reachable over the local network at the printer's IP address on port 4408. Generated gcode is uploaded through this interface rather than via raw serial or SD card.
- **Hardware:** a Creality K1-series printer modified with an Allstruder syringe pump in place of a standard extruder, printing a bacterial-alginate ink through a needle instead of extruding filament.

One architectural constraint shapes the whole queue design: agar height varies per pour, so the print height can only be measured at print time, not when a design is first queued. This is why designs sit in the queue as unprinted gcode-ready files rather than being auto-printed the moment they're submitted.

3. User Experience

The user logs into bioprinter.com and enters their passcode (only on the first visit — after that, a prompt asks whether to remain signed in on the device, and it's remembered from then on). The homepage centers on a slideshow showing, on the left, the submitted design, and on the right, a picture of the printed plate. Scrolling down reveals a timelapse of the bacteria spreading, along with photos of the

printer and lab space. Footer links point to the biopunk website, the Frontier Tower website, and the creators of the BioArtBot. Top navigation includes About, Submit a Design, Past Projects, and the biopunk logo. The homepage also shows a live queue listing the print number, the submitter's name, and the city they're printing from.

Submitting a design

Clicking “Submit a Design” opens a page similar to the homepage, but in place of the slideshow sits an upload box and a “traced image” box. Dragging and dropping an image (JPG or PNG) into the upload box replaces it with tracing options: Edge trace, Skeletonize, and Edge + Skeletonize are all complete; color fill has not been built yet and can be omitted for now.

After a trace mode is selected, the user sees a preview of the traced image (extrapolated from the generated gcode) along with controls to adjust scale and position. Below the scale adjuster, an “Add to Queue!” button opens a popup for an optional name and city, followed by a payment window. Once payment goes through, the design enters the print queue — shown under the submitter's name, or as “Print #92” if left blank — and the generated gcode is sent to the computer next to the bioprinter.

4. Hardware Setup

Connecting to the printer

Connect the printer to the same WiFi network as the computer running the bioprinter code. To reach the printer's Klipper web interface, find the printer's IP address (manually, or with the help of an LLM), enter it in a browser (note: not every browser works), and append the suffix “:4408”. For example: `http://192.168.108.60/` is the printer itself, while `http://192.168.108.60:4408/` is the Klipper portal connected to the extruder nozzle.

Installing the syringe and needle

Install the syringe and needle, filled with the material of choice. Squeeze the 3 mL syringe into the pink syringe holder, then screw on a needle. Use small screws and an allen wrench to secure the syringe holder assembly onto the printer head.

Homing

To make the printer move, first home it by clicking the “home all” button on the Klipper portal. This tells the printer where the needle tip is — it is remarkably sensitive, so there's little risk of breaking the needle (though care should still be taken not to injure yourself). The needle descends and touches the agar plate holder (currently a plate lid) multiple times during this process. Once homing is finished, raise the needle to place an agar plate in the holder.

5. Materials Preparation

Agar plates

This step can be skipped if plates are ordered pre-made. Plates must be mostly even (within about ± 0.1 mm across the surface) to avoid scratching, and must contain antibiotics matching the printed bacteria to prevent contamination. Firming plates on a flat surface works reasonably well for homemade

batches; agar volume per plate should be kept constant, and homemade plates require the agar height to be measured individually at print time.

Bacterial ink

The print medium is a 3% alginate solution mixed with 10 μL of the desired bacterial glycerol stock. The bacteria used are BL21 *E. coli*, carrying genes for fluorescent/colorful protein production and kanamycin resistance. The 3% alginate solution is highly viscous — a good property for printing, but one that makes it hard to mix glycerol stock into, tricky to draw into syringes, and hard to measure accurately. Possible fixes include adding sodium alginate to a water/glycerol-stock mixture, or placing sodium alginate inside the syringe first.

6. Print Procedure

- 1 Move the needle to position (100, 100) using the Klipper portal. Move the plate holder so the center of the plate sits directly under the needle, then place the prepared agar plate onto the holder.
- 2 Measure the height of the center of the agar by slowly lowering the needle until it grazes the plate surface, and note the z-value shown on the Klipper portal.
- 3 Change the print height in the user-submitted gcode to match this measured value.
- 4 Move the extruder plunger to meet the syringe's plunger using the Klipper portal controls. (Note: extrude and retract are swapped from what you'd expect — there must be a crossed wire somewhere.)
- 5 Optional: wet the needle with solution by extruding a small amount. Skipping this step leads to faint or nonexistent lines at the start of the print.
- 6 Move the needle above the plate. Skipping this step will bend the needle.

Once these steps are complete, upload the generated gcode to the printer through the Klipper software. The print itself takes less than a minute. Immediately after it finishes, retract the needle — some liquid remains in it and will otherwise spill onto the design.

After printing, photograph the plate and place it in a cell incubator at 37°C for 12–24 hours. For subsequent prints on an already-set-up printer, steps 1 (position) and 4 (plunger sync) can be skipped.

7. Results & Lessons Learned

The choice of a 3% alginate solution as the print medium came after a period of trial and error. Other liquids tested include distilled water, mineral oil, 3% hyaluronic acid, a 1.5% alginate solution, and bovine serum — each ruled out before settling on the current formulation.

One recurring issue was ink clinging to the outside of the needle rather than releasing cleanly onto the agar. This was resolved by switching to a higher-viscosity liquid, increasing the needle gauge, and printing as close as possible to the agar surface.

Needles currently used are BD safety-wide 22-gauge, the thinnest gauge tested so far (thinner gauges have not yet been tried). Needle tips are hand-cut at an angle to leave a clean, open hole rather than a narrowed opening. At the time of testing, only three syringes were available, making syringe count — not

alginate supply — the limiting factor for multi-color or multi-syringe prints.

Where pre-made plates weren't available, an old-school technique was used to flatten and sterilize poured plates: briefly flaming the surface with ethanol (a quick flash, immediately extinguished) while rotating the plate.

8. Current Status & Next Steps

- **Website:** the public, passcode-gated site (bioprinter.com) is the current priority, planned around Squarespace rather than custom hosting, with a collaborator (Morgan) helping connect the existing local Flask/Python backend to a hosted front end.
- **Not yet built:** color-fill tracing mode; automatic print-height measurement at queue time (blocked by the fact that agar height varies per pour and can only be measured right before printing).
- **Concentration work in progress:** preparing 4 mL LB + 3% alginate tubes, inoculating and incubating for a few hours and overnight to get more concentrated bacterial samples; a plan for a high-concentration glycerol stock (centrifuging ~10 mL of overnight culture down to a pellet, resuspending into ~1.5 mL, and freezing immediately at -80°C) is also in progress.
- **Versioning:** the bioprinter code and protocols are kept on GitHub so the project doesn't depend on any one person.

Photos, diagrams, and queue/website screenshots to be added before final distribution.